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TEXAS
The University of Texas at Austin

THE MEMORY SUBSYSTEM

C S 350C (Advanced Computer Architecture)

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Sources:

- "HIGH-PERFORMANCE DRAM SYSTEM DESIGN CONSTRAINTS AND CONSIDERATIONS", MS thesis, Joseph Gross, University of Maryland, College Park, 2010.
- Balasubramonian, Rajeev. Innovations in the Memory System, Springer, 2019.
- S. Rixner, W. Dally, U. Kapasi, P. Mattson, and J. Owens. Memory access scheduling. In Proc. of ISCA, 2000.
- Smruti R. Sarangi. Next-Gen Computer Architecture (1st edition), White Falcon, 2023. ISBN 8119510143.
- Vijay Nagarajan, Daniel J. Sorin, Mark D. Hill, and David A. Wood. A Primer on Memory Consistency and Cache Coherence, Second Edition. Synthesis Lectures on Computer Architecture #49.

1. The Device Level

1T1C

Sources:

- "HIGH-PERFORMANCE DRAM SYSTEM DESIGN CONSTRAINTS AND CONSIDERATIONS", MS thesis, Joseph Gross, University of Maryland, College Park, 2010.

Organization of DRAM Device

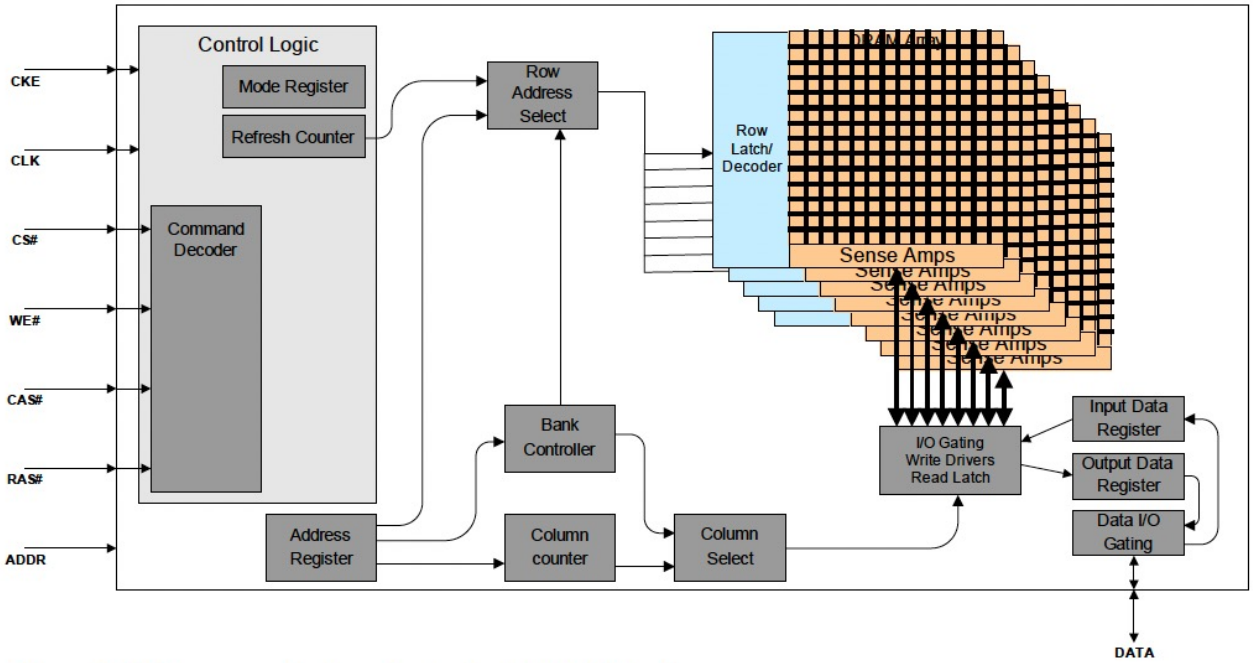


Figure 2.1: The organization of a typical DRAM device

Storing One Bit of Data

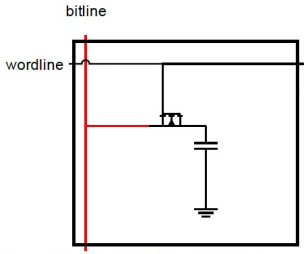
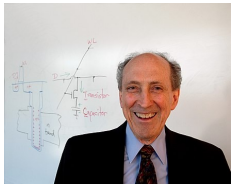


Figure 2.2: The 1T1C Storage Cell

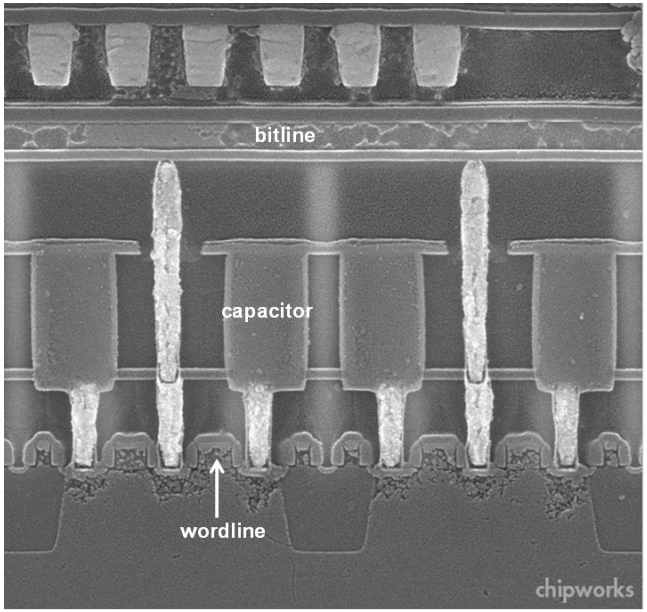
- 1T1C cell
  - United States Patent #3,387,286, *Field-Effect Transistor Memory*, 1968-06-04.
  - Inventor: Robert H. Dennard (1932-2024).
  - Link to patent PDF.



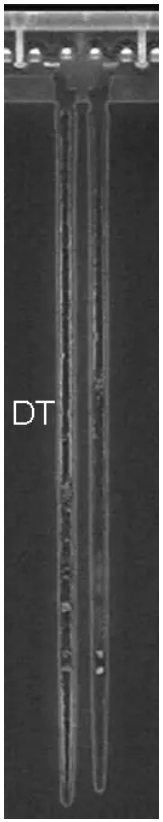
- Cell capacitance is ~20 fF.
  - Very small.
- Charged capacitor will leak and must therefore be periodically refreshed.
  - Refresh interval t_{REFI} depends on temperature, since leakage rate is directly proportional to temperature.

Capacitor Types

- Planar
  - Large footprint, costly
- Stacked
  - Relatively smaller footprint
- Deep Trench
  - Standard CMOS process
  - Very small footprints, hence very high density
  - Below substrate, hence ease of fabrication

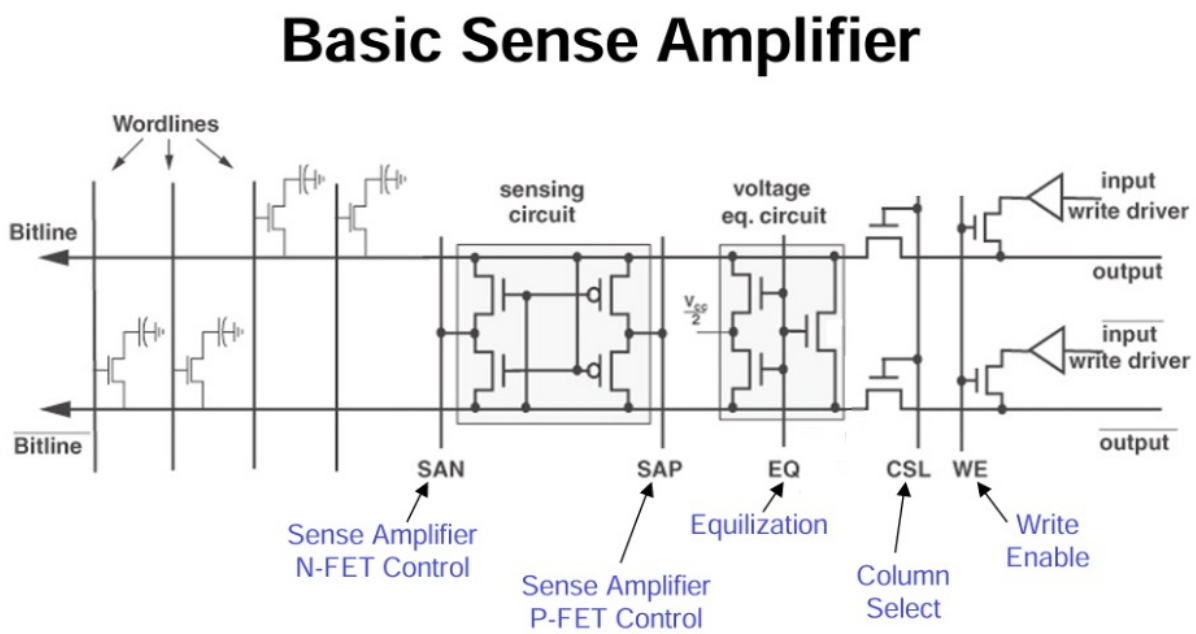


Stacked Capacitor



Deep Trench capacitor

Sensing and Amplifying Voltages



Sense Amplifiers

- The *sense* circuit
  - Two nMOS and two pMOS devices.
    - nMOS conducts when $V_{gs} > V_{th}$; pMOS when $V_{gs} < V_{th}$.
    - For nMOS, $V_{th} > 0$; for pMOS, $V_{th} < 0$.
  - Init:
    - Bitlines are precharged to $V_{DD}/2$ (using the voltage equalization circuit).
    - $SAN = 0, SAP = V_{DD}$.
  - When capacitor is connected, its value (V_{DD} or 0) will slightly alter the charge on the line, causing either pMOS (for V_{DD}) or nMOS (for 0) to conduct.
  - Eventually, the bitlines will be driven to V_{DD} and 0, thus amplifying the capacitor's charge.

- The *voltage equalization* circuit
  - Three nMOS devices.
  - If $EQ = 0$:
    - All devices are non-conducting.
    - Both bitlines are floating.
  - If $EQ = V_{DD}$:
    - All devices are conducting.
    - Both bitlines are set to $V_{DD}/2$.

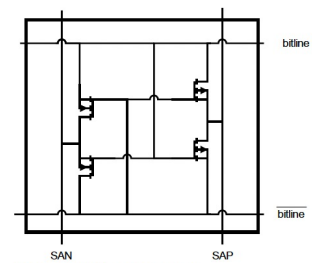


Figure 2.3: The sense circuit

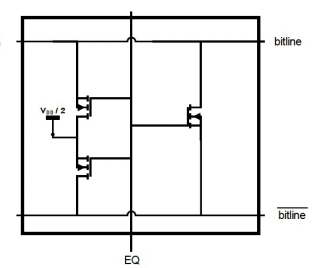


Figure 2.4: Voltage equalization circuit

Access Steps in DRAM Device

1. Sense bit values of cells in a row (aka “open a row”).
2. Restore values. Perform reads or writes on the opened row.
3. Precharge sense amplifiers (aka “close the row”).

Step 1: Sense

- $EQ = 0$.
  - Bitlines are floating (at what value?).
- Wordline activated to allow capacitor's charge to influence upper bitline.
  - Ex: Cap holds 1, so bitline voltage increases.
- Starts a feedback loop that eventually drives upper bitline to V_{DD} and lower bitline to 0.
- If cap held 0, bitlines would have moved the other way.
- In either case, the upper and lower bitlines end up at opposite values, correctly representing the value stored in the capacitor.

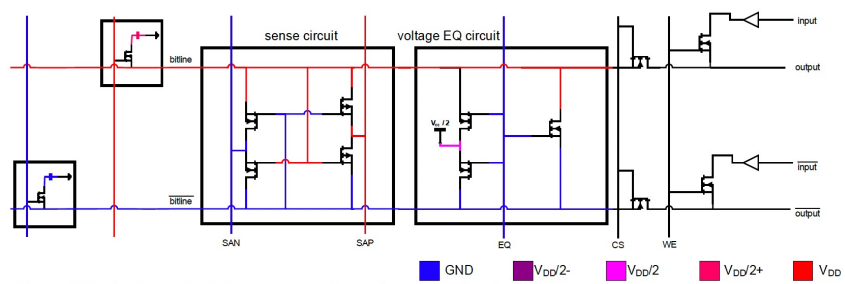


Figure 2.6: At the end of the sense portion of a row activation

Step 2a: Restore, Read

- Activate chip select (CS) to connect the amplified value to the output.
- Restore the value from the capacitor to its original state.
  - Keep the wordline active and maintain the voltages on the bitlines for a minimum period of time (t_{RAS}).
  - After this time, the row can be closed with a precharge operation.
- Need to wait for values to stabilize on bitlines before reading or writing.
  - This parameter is called t_{RCD}.

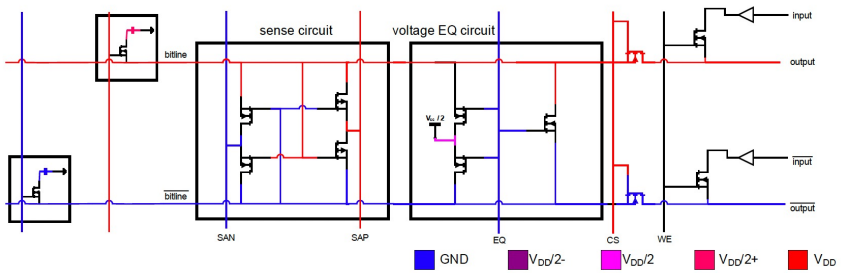


Figure 2.7: The restore and read portion a row activation

Step 2b: Write to An Active Row

- Wordline and CS stay asserted.
- Write enable (WE) is also asserted.
  - This activates the transistor that controls access to the write buffer of the bitline.
  - The $input$ and $\overline{input}$ signals update the stored value in the cell.
  - Writing takes additional time t_{WR} to complete.
  - After this time, the row may be closed (precharged).

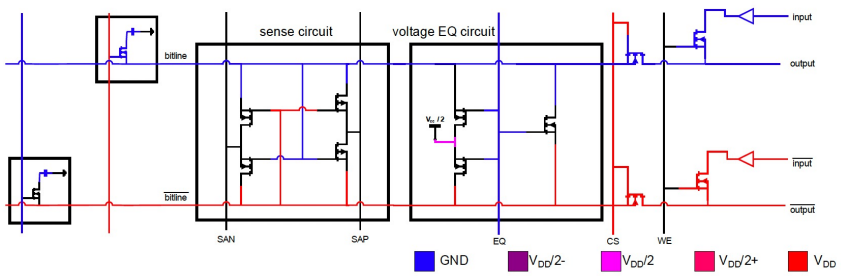


Figure 2.8: A write to an active row

Step 3: Precharge a Row

- Needs to happen between row activations.
- CS, WE, wordline are all de-asserted.
- EQ is active.
- This sets both bitlines to $V_{DD}/2$.
  - This is a *metastable* condition that will be perturbed by the small charge from the capacitor and amplified using the feedback loop from step 1.
- The row precharge time t_{RP} specifies how much time needs to pass for the bitlines to be set to $V_{DD}/2$.

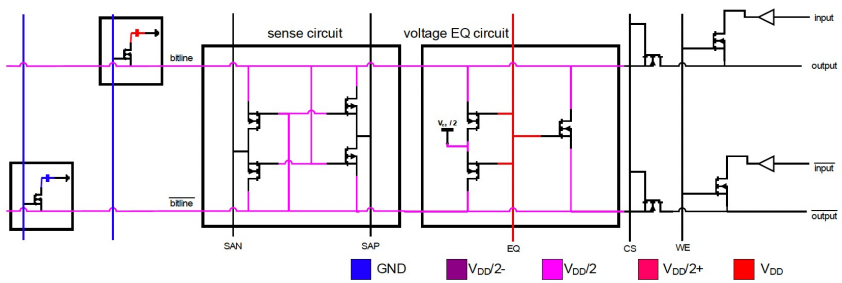


Figure 2.9: Precharging a row in preparation for a row activation